Comprehensive Phytochemical Characterisation of *Funtumia Elastica* Root Ethanolic Extracts: Exploring Potentials for Male Fertility Enhancement

[*1]Eseine-Aloja C.E., [2]Ugbeni O.C., [2]Osubor C.C., [2]Akpeh P.K.

[1]Department of Medical Biochemistry, Faculty of Basic Medical Sciences, Ambrose Alli University, Ekpoma, Edo State, Nigeria
[2]Department of Biochemistry, Faculty of Life Sciences, University of Benin, Benin City, Nigeria

ABSTRACT

Traditional medicinal plants, such as *Funtumia elastica* (*F. elastica*), are used ethnomedicinally to enhance male reproductive health, yet scientific evidence regarding their phytochemical composition remains limited. This study aimed to profile the phytochemical constituents of *F. elastica* root ethanolic extracts (FEREE) and their potential implications for male fertility enhancement. Roots of *F. elastica* were collected, authenticated, and extracted using 40% ethanol. The percentage yield was determined, followed by qualitative and quantitative phytochemical screening. High-performance liquid chromatography (HPLC) was used to identify and quantify bioactive compounds. The percentage yield of FEREE was 5.38%. Qualitative analysis revealed the presence of coumarins, flavonoids, cardiac glycosides, steroids, and terpenoids. Quantitative evaluation showed concentrations of coumarins (0.064 ± 0.005 mg/ml), flavonoids (0.303 ± 0.009 mg/ml), cardiac glycosides (0.554 ± 0.004 mg/ml), steroids (0.161 ± 0.009 mg/ml), and terpenoids (0.241 ± 0.007 mg/ml). HPLC analysis identified key bioactive compounds including quercetin, funtumine, kaempferol, hydrobenzoic acid, and roemerine. Many of these compounds possess antioxidant, anti-inflammatory, and endocrine-modulatory properties that may protect spermatogenic cells, improve sperm quality, and support hormonal balance. *F. elastica* roots are rich in bioactive phytochemicals with potential applications in male fertility enhancement.

Keywords: *Funtumia elastica*, Male fertility, Phytochemical, Quercetin, Reproductive health.

***Correspondance: Email:** cee-aloja@aauekpoma.edu.ng, **Phone number:** +2348037950022

1. INTRODUCTION

Male infertility remains a significant global reproductive health challenge, contributing to nearly half of all infertility cases among couples [1, 2]. Oxidative stress, hormonal imbalance, and disruptions in spermatogenesis are well-recognized mechanisms underlying impaired male reproductive function [3]. Although several conventional therapies exist, they often face limitations such as high cost, inconsistent efficacy, limited accessibility, and undesirable side effects. These constraints have stimulated growing scientific interest in medicinal plants as alternative or complementary therapeutic agents [4]. Many plants used in traditional medicine possess rich profiles of bioactive compounds including flavonoids, alkaloids, saponins, and terpenoids known for their antioxidant and endocrine-modulatory activities. Consequently, ethnomedicinal remedies, particularly those employed in African traditional medicine, offer promising avenues for discovery due to their phytochemical diversity and long-standing historical use [5].

Funtumia elastica, commonly known as the "silk rubber tree," belongs to the *Apocynaceae* family and is widely distributed across West and Central Africa. It holds a long history of ethnomedicinal use, having been traditionally employed for various conditions including infections, inflammation, systemic ailments,

and reproductive concerns [6]. Studies on members of the Apocynaceae family indicate that they are rich in pharmacologically active phytochemicals such as alkaloids, flavonoids, terpenoids, and saponins with demonstrated antioxidant, anti-inflammatory, and endocrine-modulating properties [7, 8]. These characteristics position *F. elastica* as a plant of interest in the search for natural agents capable of mitigating factors that contribute to male infertility.

Despite its diverse ethnomedicinal applications, scientific evidence on the reproductive health potentials of *F. elastica* remains limited, as this area has been largely underexplored in rigorous reproductive biology research [9, 10]. Specifically, there is a paucity of data concerning the phytochemical constituents of the plant particularly its root extracts and how these phytochemicals might influence male fertility. Existing studies have focused primarily on antimicrobial or general phytochemical properties, leaving notable gaps regarding its reproductive applications [11]. Understanding the phytochemical composition of its root extracts is therefore essential for elucidating possible mechanisms by which the plant may influence reproductive outcomes, especially given the established roles of plant-derived alkaloids, flavonoids, and saponins in mitigating oxidative stress and supporting male reproductive function [12].

Flowing from the above, this study seeks to bridge these gaps by profiling the phytochemical constituents of ethanolic root extracts of *F. elastica* and exploring their potential implications for male fertility enhancement and overall male reproductive health. By providing a detailed phytochemical evaluation alongside fertility-focused considerations, this research will contribute to the growing body of work supporting plant-based interventions for male reproductive health, while offering a scientific foundation for future pharmacological investigations and therapeutic development. Furthermore, identifying the specific compounds present in the root extract may serve as a guide for subsequent *in vitro* or *in vivo* studies on male fertility.

2. MATERIALS AND METHODS

Plant Root Collection and Identification: Fresh roots of *Funtumia elastica* were collected from Ebhoran in Etsako West Local Government Area, Edo State, Nigeria. The plant material was authenticated in the Department of Botany, Ambrose Alli University, Ekpoma, Edo State, Nigeria. A voucher specimen of *Funtumia elastica* was deposited at the herbarium of the same department and assigned the voucher identification number 1045 for future reference.

Preparation of Plant Material and Extraction: The collected roots of *Funtumia elastica* were thoroughly washed to remove soil and other debris and subsequently air-dried at room temperature. The dried roots were pulverized into coarse powder using a mechanical grinder. The powdered root material was macerated in 40% ethanol and kept on a mechanical shaker at room temperature for 72 hours to facilitate extraction [13]. Following extraction, the mixture was filtered using Whatman No. 1 filter paper into a clean flask. The filtrate obtained was concentrated using a rotary evaporator at 40 °C to remove the solvent. The concentrated extract was subsequently subjected to freeze-drying (lyophilization) to obtain the dried powdered extract. The resulting *Funtumia elastica* root ethanolic extract (FEREE) was stored in airtight containers and refrigerated until further analysis [14].

Qualitative Phytochemical Screening: Qualitative phytochemical screening of the ethanolic root extract of *F. elastica* was carried out using standard methods described by Harborne [15], Trease and Evans [16], and Sofowora [17]**.** The extract was tested for the presence of major phytochemical classes such as tannins, coumarins, flavonoids, cardiac glycosides (via Keller Killiani test), saponins,

steroids, terpenoids (via Salkowski test), phenols, phlobatannins, alkaloids (via Mayer's test), and anthraquinones.

Quantitative Phytochemical Determination: The concentration of phytochemicals were determined qualitatively as follows; Coumarin concentration was determined using Esculin as the calibration standard according to the method described by Fatima *et al.* [18]**.** Total flavonoid content was determined using the aluminium chloride colorimetric method with quercetin as standard as described by Agarwal *et al.* [19]**.** Cardiac glycoside concentration was determined using securidaside as standard according to the method of Tofighi *et al.* [20]**.** Steroid concentration was determined using stigmasterol as the standard following the method described by Agarwal *et al.* [19]**.** Terpenoid concentration was determined using β-sitosterol as standard following the vanillin-sulfuric acid method described by Pedrosa *et al.* [20]. Total phenolic content was determined using the Folin–Ciocalteu method as described by Tofighi *et al.* [21] with gallic acid as the standard.

Identification and Quantification of Bioactive Compounds by High-Performance Liquid Chromatography (HPLC): Identification and quantification of bioactive compounds present in the ethanolic root extract of *Funtumia elastica* were performed using HPLC according to the method described by Kaur *et al.* [22]. The procedure involved the following steps: Sample Preparation: The dried extract was dissolved in HPLC-grade solvent and filtered through a 0.45 µm membrane filter before injection. Chromatographic Separation: Separation of phytochemical constituents was achieved using an HPLC system equipped with a reverse-phase column, under optimized mobile phase conditions. Detection: Detection of separated compounds was carried out using a UV-visible detector at appropriate wavelengths suitable for phenolic and flavonoid compounds.

Data Analysis: Chromatographic peaks were analyzed using specialized chromatography software. Identification of compounds was performed by comparing retention times and spectral characteristics with those of authentic reference standards. Quantification of identified phytochemicals was carried out based on calibration curves generated from standard compounds.

3. RESULTS

Qualitative Phytochemical Composition of *Funtumia elastica* Root Ethanolic Extract (FEREE)

The qualitative phytochemical screening of the ethanolic root extract of Funtumia elastica revealed the presence of several secondary metabolites. The results indicated that coumarins, flavonoids, cardiac glycosides, steroids, and terpenoids were present in the extract, whereas tannins, saponins, phenols, phlobatannins, alkaloids, and anthraquinones were not detected. Cardiac glycosides showed the highest level of occurrence, while flavonoids, steroids, and terpenoids were present at relatively high levels. The qualitative phytochemical profile of the extract is presented in Table 1.

Quantitative Phytochemical Composition of FEREE

Quantitative phytochemical analysis further confirmed the presence of several bioactive constituents in the ethanolic root extract. The results revealed that cardiac glycosides had the highest concentration (0.554 ± 0.004 mg/ml), followed by flavonoids (0.303 ± 0.009 mg/ml) and terpenoids (0.241 ± 0.007 mg/ml). Steroids were present in moderate concentration **(**0.161 ± 0.009 mg/ml), while coumarins were detected in relatively low concentration (0.064 ± 0.005 mg/ml). The quantitative phytochemical composition of FEREE is presented in Table 2.

Identification and Quantification of Bioactive Compounds in FEREE by High-Performance Liquid Chromatography (HPLC)

The identification and quantification of bioactive compounds present in the ethanolic root extract of *Funtumia elastica* were carried out using HPLC. The chromatographic profile revealed the presence of several phenolic compounds, flavonoids, alkaloids, and other phytoconstituents within the extract. The chromatogram illustrating the separation and detection of these compounds is presented in Figure 1.

Table 1: Qualitative phytochemical composition of FEREE

S/N	Phytochemicals	FEREE
1	Tannins	–
2	Coumarin	+
3	Flavonoids	++
4	Cardiac glycosides	+++
5	Saponins	–
6	Steroids	++
7	Terpenoids	++
8	Phenols	–
9	Phlobatannins	–
10	Alkaloids	–
11	Anthraquinones	–

Key: – = Not detected, + = Present (low), ++ = Present (high), +++ = Present (very high)

Table 2: Quantitative phytochemical composition of FEREE

S/N	Phytochemicals	Unit	FEREE
1	Coumarin	mg/ml	0.064 ± 0.005
2	Flavonoids		0.303 ± 0.009
3	Cardiac glycosides		0.554 ± 0.004
4	Steroids		0.161 ± 0.009
5	Terpenoids		0.241 ± 0.007

Analysis of the chromatographic peaks enabled the identification of twenty (20) bioactive compounds based on their retention characteristics and comparison with reference standards. Among the identified compounds were hydroxybenzoic acid, gallic acid, caffeic acid, catechin, epicatechin, resveratrol,

quercetin, and kaempferol, which are well-known phenolic and flavonoid compounds with established antioxidant and pharmacological activities. Additionally, several alkaloids and other bioactive molecules such as funtumine, funtumidine, vindoline, vinblastine, ajmalicine, and echitovenine were also detected in the extract. Notably, quercetin exhibited the highest peak area (8057.22), indicating that it was the most abundant compound detected in the chromatographic profile of the extract. This was followed by kaempferol (2293.82) and funtumine (2371.64), suggesting that flavonoids and alkaloid-type compounds constitute significant components of the extract. The detailed characteristics of the identified compounds, including their molecular formulas, molecular weights, and chromatographic peak areas, are presented in Table 3

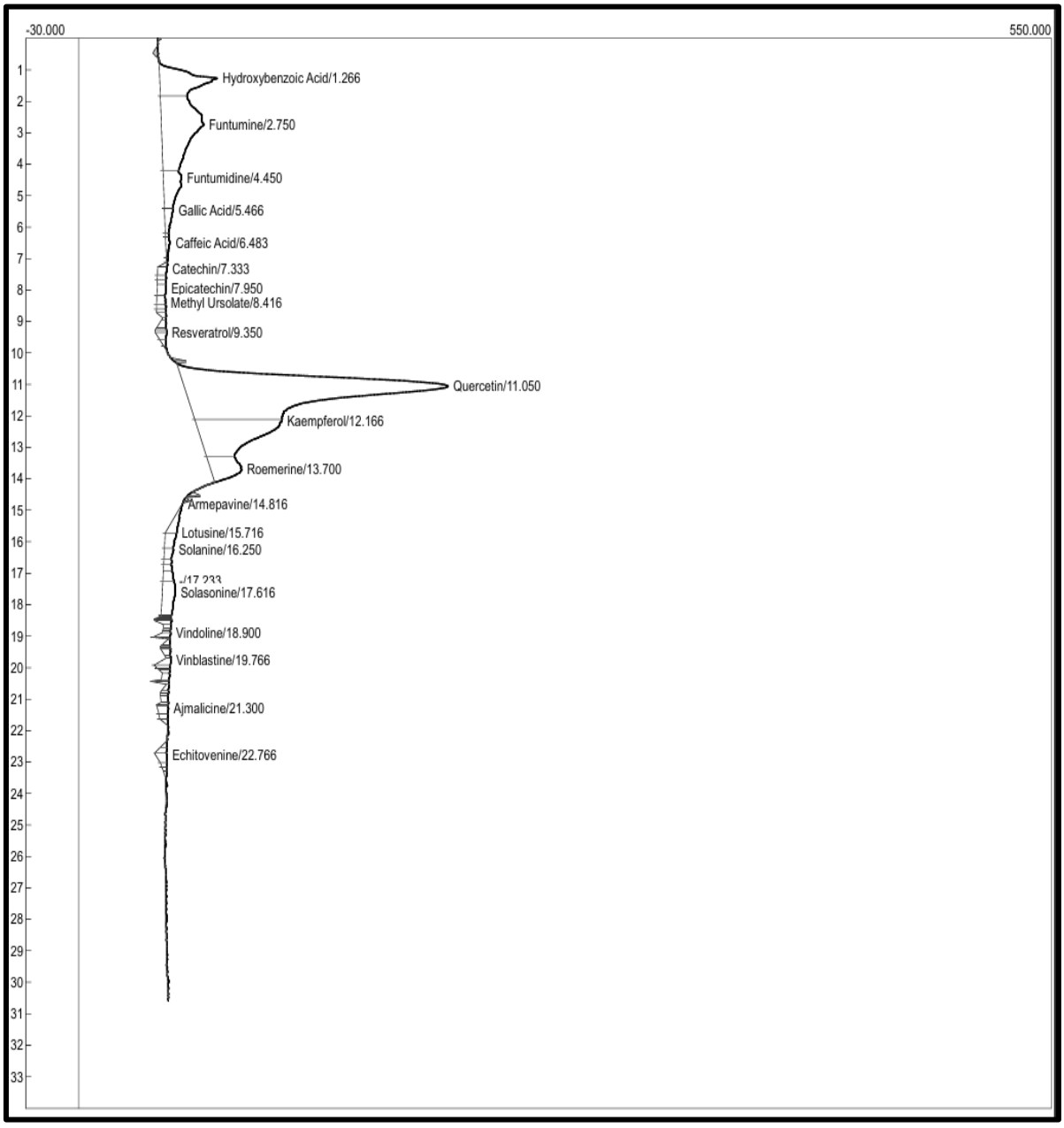


Figure 1: HPLC chromatogram showing the identification and quantification of bioactive compounds in the ethanolic root extract of *Funtumia elastica*.

4. DISCUSSION

The present study provides a comprehensive phytochemical characterization of the ethanolic root extract of *Funtumia elastica* through qualitative screening, quantitative estimation, and high-performance liquid chromatography (HPLC) profiling. The results revealed the presence of several biologically important secondary metabolites, including coumarins, flavonoids, cardiac glycosides, steroids, and terpenoids. The detection of these phytochemical classes, alongside numerous specific bioactive compounds identified by chromatographic analysis, provides scientific insight into the chemical composition of *F. elastica* root and supports its traditional medicinal use. Variations in phytochemical composition among plant species and plant parts are well documented and are influenced by factors such as geographical conditions, harvesting season, extraction methods, solvent polarity, and the physiological status of the plant [23, 24].

Qualitative phytochemical screening confirmed the presence of coumarins, flavonoids, cardiac glycosides, steroids, and terpenoids, while tannins, saponins, phenols, phlobatannins, alkaloids, and anthraquinones were not detected using the applied screening methods. Secondary metabolites such as flavonoids and terpenoids are widely recognized for their antioxidant and anti-inflammatory properties, which play essential roles in protecting tissues from oxidative damage [10, 25]. These properties are particularly relevant to male reproductive physiology because oxidative stress is a major contributor to impaired spermatogenesis, reduced sperm motility, and DNA damage in spermatozoa..

Table 3: **Characteristics of bioactive compounds identified in FEREE by HPLC**

SN	Phytochemical	Formula	Molecular Weight (g/mol)	Area Covered
1	Hydroxybenzoic acid	$C_7H_6O_3$	138.12	1114.26
2	Funtumine	$C_{21}H_{35}NO$	317.50	2371.64
3	Funtumidine	$C_{21}H_{37}NO$	246.00	570.25
4	Gallic acid	$C_7H_6O_5$	170.12	163.92
5	Caffeic acid	$C_9H_8O_4$	180.16	63.50
6	Catechin	$C_{15}H_{14}O_6$	290.26	82.00
7	Epicatechin	$C_{15}H_{14}O_6$	290.27	102.31
8	Methyl urosolate	$C_{31}H_{50}O_3$	470.70	82.31
9	Resveratrol	$C_{14}H_{12}O_3$	228.25	69.30
10	Quercetin	$C_{15}H_{10}O_7$	302.24	8057.22
11	Kaempferol	$C_{15}H_{10}O_6$	286.23	2293.82
12	Roemerine	$C_{18}H_{17}NO_2$	279.30	671.32
13	Armepavine	$C_{19}H_{23}NO_3$	313.39	175.34
14	Lotusine	$C_{19}H_{24}NO_3$	314.40	164.74
15	Solanine	$C_{45}H_{73}NO_{15}$	868.06	103.28
16	Solasonine	$C_{45}H_{73}NO_{16}$	884.10	435.29
17	Vindoline	$C_{25}H_{32}N_2O_6$	465.53	65.24
18	Vinblastine	$C_{46}H_{58}N_4O_9$	810.97	77.79
19	Ajmalicine	$C_{21}H_{24}N_2O_3$	352.43	81.93
20	Echitovenine	$C_{23}H_{28}N_2O_4$	396.50	95.64

Quantitative phytochemical analysis revealed that cardiac glycosides were present in the highest concentration in the extract, followed by flavonoids and terpenoids. The relatively high abundance of flavonoids is of particular importance in the context of male fertility because flavonoids have been widely reported to improve sperm quality by scavenging reactive oxygen species, modulating inflammatory pathways, and supporting endocrine function involved in testosterone production and spermatogenesis [26]. Steroidal compounds detected in the extract may also contribute to reproductive health because plant sterols serve as precursors for steroid hormone synthesis and play roles in maintaining membrane stability in sperm cells [27]. Terpenoids are similarly recognized for their antioxidative and endocrine-modulating activities, with some acting as phytoandrogenic compounds that may support testosterone biosynthesis and sperm maturation [28].

Although cardiac glycosides are primarily known for their cardiotonic effects through inhibition of the sodium-potassium ATPase pump, emerging evidence suggests that at low concentrations they may influence calcium signaling pathways that are important for sperm motility and flagellar activity [29, 30]. However, excessive exposure to cardiac glycosides can produce toxic effects, including potential reproductive toxicity. This highlights the importance of dose optimization when considering plant extracts containing these compounds for therapeutic applications [31, 32].

HPLC profiling further revealed a diverse chemical composition in the root extract, with twenty phytochemicals identified, including phenolic acids, flavonoids, and alkaloid-type compounds. Notable compounds included hydroxybenzoic acid, gallic acid, caffeic acid, catechin, epicatechin, resveratrol, quercetin, and kaempferol, all of which are well-known antioxidants capable of protecting biological systems against oxidative damage by scavenging reactive oxygen species and enhancing cellular antioxidant defense mechanisms [33]. Among these compounds, quercetin exhibited the highest peak area, indicating that it is a major constituent of the extract.

Quercetin is one of the most extensively studied flavonoids with respect to male reproductive health. It has been reported to improve sperm count, motility, morphology, and DNA integrity while protecting Leydig and Sertoli cells from oxidative and inflammatory damage [34, 35]. Quercetin has also been shown to enhance testosterone production through modulation of Leydig cell steroidogenesis and to protect against drug-induced reproductive toxicity [36, 37]. Similarly, kaempferol possesses strong antioxidant and anti-apoptotic properties and has been shown to improve spermatogenic function and preserve the steroidogenic capacity of testicular cells [38, 39]. Hydroxybenzoic acid and related phenolic derivatives such as gallic acid also exhibit protective effects against oxidative injury in reproductive tissues and have been associated with improvements in sperm quality parameters [40].

In addition to phenolic compounds, the chromatographic analysis detected several indole-type alkaloids, including funtumine, funtumidine, vindoline, vinblastine, ajmalicine, and echitovenine. Alkaloids represent an important group of plant secondary metabolites with diverse pharmacological activities. Funtumine, a characteristic alkaloid of *F. elastica*, has previously demonstrated cytotoxic activity against cancer cells [41]. Although its direct role in male reproductive physiology remains poorly understood, its antiproliferative properties suggest that caution may be necessary, as rapidly dividing germ cells within the seminiferous epithelium could be sensitive to cytotoxic compounds. Roemerine, another alkaloid detected in the extract, has been reported to possess antimicrobial and anti-inflammatory activities, although its

reproductive effects have not been well studied.

An important observation from the present study is that most previous phytochemical investigations of *F. elastica* have focused primarily on the leaves, latex, or stem bark of the plant, whereas the root has received comparatively little scientific attention. Because different plant organs synthesize and accumulate metabolites for specific physiological functions, variations in phytochemical composition among plant parts are expected. The present findings therefore contribute valuable new information regarding the chemical composition of *F. elastica* root and expand current knowledge of its bioactive constituents.

Overall, the results of this study demonstrate that the ethanolic root extract of *F. elastica* contains a complex mixture of bioactive phytochemicals with antioxidant, anti-inflammatory, and endocrine-modulating properties. The predominance of compounds such as quercetin, kaempferol, and hydroxybenzoic acid, which have well-documented protective roles in male reproductive physiology, suggests that the extract may possess potential for supporting male fertility through mitigation of oxidative stress and modulation of steroidogenic pathways. However, the presence of bioactive alkaloids and cardiac glycosides also highlights the importance of careful dose evaluation and toxicological assessment. Further in vivo and mechanistic studies are therefore required to fully elucidate the reproductive effects of *F. elastica* root extract and to determine its safety and efficacy as a potential phytotherapeutic agent for male fertility enhancement.

5. CONCLUSION

This study provides a comprehensive phytochemical characterization of the ethanolic root extract of *Funtumia elastica* using qualitative, quantitative, and HPLC analyses. Many of the identified compounds, particularly flavonoids and phenolic constituents such as quercetin and kaempferol, are known for their antioxidant and anti-inflammatory properties, which play important roles in protecting reproductive tissues from oxidative stress and supporting normal spermatogenic and endocrine functions. These findings therefore highlight the potential relevance of *F. elastica* root extract to male reproductive health and suggest possible fertility-enhancing properties. Furthermore, this study contributes valuable information to the existing knowledge of the phytochemical composition of *F. elastica*, particularly the root, which has received comparatively less scientific attention than other parts of the plant. The results provide scientific support for the ethnomedicinal use of the plant and underscore its potential as a source of bioactive compounds for future therapeutic development. Overall, the findings provide a strong basis for further pharmacological and reproductive biology investigations, including detailed *in-vitro* and *in-vivo* studies, to elucidate the mechanisms of action, safety profile, and therapeutic potential of *F. elastica* root extract in relation to male fertility enhancement.

Further studies should be conducted to evaluate the biological effects of *Funtumia elastica* root extract on male reproductive function using well-designed *in vivo* and *in vitro* experimental models. Such studies should focus on assessing its effects on spermatogenesis, sperm quality parameters, and reproductive hormone profiles in order to clarify its potential fertility-enhancing properties. Additionally, toxicological and dose–response studies are necessary to establish the safety profile of the extract, particularly considering the presence of bioactive compounds such as cardiac glycosides and alkaloids. Future research should also aim to isolate and characterize the individual phytochemicals responsible for the observed biological activities and investigate their mechanisms of action at the molecular level. Ultimately, such studies may facilitate

the development of standardized phytotherapeutic formulations derived from *F. elastica* for potential application in the management of male reproductive disorders.

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CONFLICT OF INTEREST

The authors declare no conflict of interest

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